Vortex86DX
Layout Guideline

Ver. 0.2

Date (mm/dd/yy)	Version	Description
06/27/08	0.1	Initial Release
08/20/08	0.2	Add DDR2's power and ground via pattern

Overview

1.1. General Description

The following document provides some technical data that describe layout guideline of VORTEX86DX. The main purpose of this document was to provide VORTEX86DX with high quality signal for system reliable operation.

1.2. PCI bus Layout Guideline

Control the skew of PCI clock between VORTEX86DX and PCI device within 2ns. Routing signals of PCI with 5mil trace width and keep minimum 10mil space. Add a 100ohm resistor on each IDEVSEL input of PCI devices. Route PCI trace length within 8 inch. If application need to combine PCI slot and down devices. Place down devices on PCI bus end to improve signal integrity. DMP's PCI interface isn't support 5V tolerance. Please don't connect to PCI device of 5V. Some signals need to pull up or pull down for the interface that we recommend using 10k ohm resistors.

1.3. IDE bus Layout Guideline

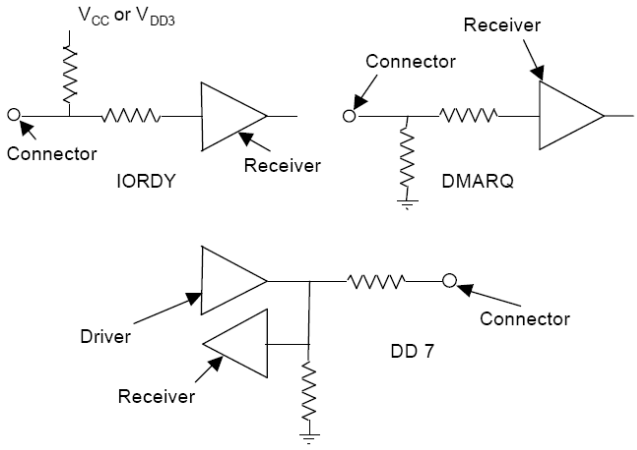
Matches trace length of IOR- and IORDY within 100mil. Matches trace length of IOR-/IORDY and D0~D15 within 400mil. Control the trace length of IDE interface within 6 inch. Routing signals of IDE with 5mil trace width and keep minimum 10mil space. We recommend to buffer DASP- signal to driver LED to reduce EMI issue. The pull up and pull down is required for interface on host side as follows:

1. The DMARQ needs a 5.6k ohm pull down resistor.
2. The IORDY needs a 4.7k ohm pull up resistor.
3. The INTRQ needs a 10k ohm pull down resistor.
4. The CSEL connect to ground.
5. DASP- needs a 100k ohm pull up resistor if buffer the signal to driver LED.

The series termination required for the interface on host side as follows:

1. The IOR-, IOW- and DMACK- need a 22ohm series termination resistor.
2. The CS0-, CS1-, DA0~2, D0~15 and RESET- need a 33ohm series termination resistor.
3. The DMARQ, INTRQ and IORDY need an 82ohm series termination resistor.

The topology for series termination with pull up or pull down is as follows:



Place all series termination near IDE connector.

1.4. Ethernet PHY Layout Guideline

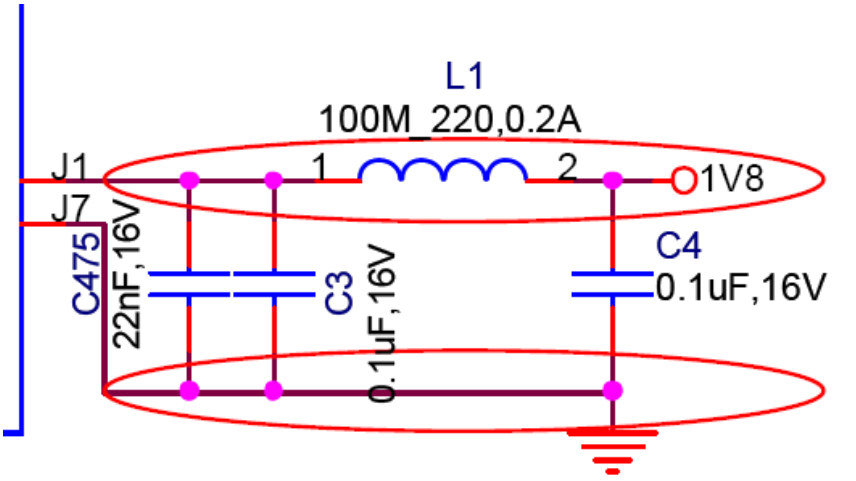
Place termination resistor near IC side. Route differential pair of Ethernet PHY with 5.5mil traces width and 9.5mil space between TXP (RXP) and TXN (RXN), keep minimum 30 mils space for other traces to reduce crosstalk and EMI as possible. Keep the analog signals away from digital signals. Place ISET resistor near C6032 at least with trace width of 20mil. Place Transform near RJ45 connector and place termination resistor near IC side.

1.5. USB PHY Layout Guideline

Route differential pair of USB PHY with 7mil traces width and 8mil space between DP and DN, keep minimum 30 mils space for other traces to reduce crosstalk and EMI as possible. Keep the analog signals away from digital signals. Place ISET resistor near C6032 at least with trace width of 20mil. Place common chock near connector if present.

1.6. PLL layout Guideline

A clear power supply is importance for PLL in high performance CPU. DMP recommend a filter technology to reduce jitter as follows:



1.7. SD Card Interface Layout Guideline

Route traces length of SD Card Interface within 1000mil.

1.8. DDR2 Interface layout Guideline

More attention must be paid to high speed rule for board layout and bus loading to make sure that short clock cycle time can be met without noise, ringing, crosstalk power or ground bounce. To provide a good return path and limit above noise on the DDR interface, all system memory signals should reference a single contiguous plane as possible.

The recommendation layout guideline for a DDR device is as follows:

Please connect with DMP if the application needs more than one DDR device.

1. In order to attain highest speed as possible. We suggest the signals of DQS/DQS#, DQM and DQ only a loading preference. Please connect with DMP if the signals of DQS/DQS#, DQM and DQ need to load more than one.
2. The DDR clock is differential signal that need a 100 ohm differential terminal resistor near DDR side for a DDR configuration or on split point for two DDRs configuration, and keep the trace length under 0.2 inch from split point to respective DDR. Reserve location of two 33ohm damping resistors near CPU side for low power application. The two trace width is 5 mil and space 5mil to get near 100ohm differential impedance, and keep minimum 15 mils space for other traces to reduce crosstalk as possible. The trace length of two differential signals must be match within 10mil. The category signals were routed within 1.4 inch.
3. Routing each signals of byte lane on same layer with same via count. Control the number of via within 2. The mismatch of trace length must be within 20mil for each signals of byte lane. The high byte lane is DQS1/DQS1#, DQM1 and DQ8~DQ9 and the low byte lane is DQS0/DQS0#, DQM0 and DQ0~DQ7

respective. Place a damping resistor on middle of each signal to trade off the timing of read and write for low power with ODT off. The DQS/DQS# is differential signal that they were routed differential with 5mil trace width and 5mil space between DQS and DQS#, keep minimum 15 mils space for other traces to reduce crosstalk as possible. The other signals were routed 5mil trace width and keep minimum 10mil space between signals to reduce crosstalk and control impedance of trace. The high byte lane and low byte lane must be matching within 0.2 inch. The category signals were routed within 1.2 inch. It is importance to route the differential signals of DQS/DQS# within 10mil.

4. Routing signal of command and address with a 33ohm damping resistor near CPU side. The technology of VTT terminal was prohibited since the address pins of DMP support bootstrap function. The category signals were routed 5mil trace width with minimum 10mil space between signals to reduce crosstalk and control impedance of trace. The signals of command and address must be matching within 0.1 inch. The category signals were routed within 1.4 inch.

5. To provide a good return path, all system memory signals should reference a single contiguous plane as possible.

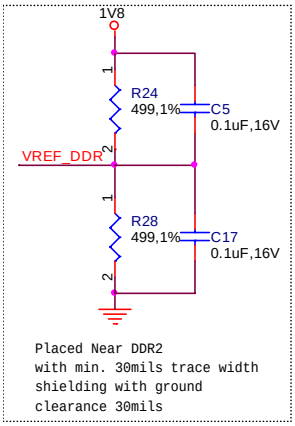
6. DDR memory signal trace edges must be at least 4 times (20 mils) trace width from any non-DDR memory signal trace edge.

7. Routing trace near edge of the reference plane should keep minimum 5 times trace width from edge of signal trace to edge of the reference plane.

8. The signals of address and command must be following Balanced-T routing rule for the application of two DDR devices.

9. The generation of Vref is critical. It must be follow the variance of VDDQ. We are suggestion to generate the reference from VDDQ by two 499ohm divide resistors with parallel 0.1uf X7R type bypass capacitor for each resistor. Place the divide resistors and bypass capacitor on quiet area. Also place a 0.1uf X7R type bypass capacitor on Vref pin of each device.

10. The circuit of Vref is as follows:



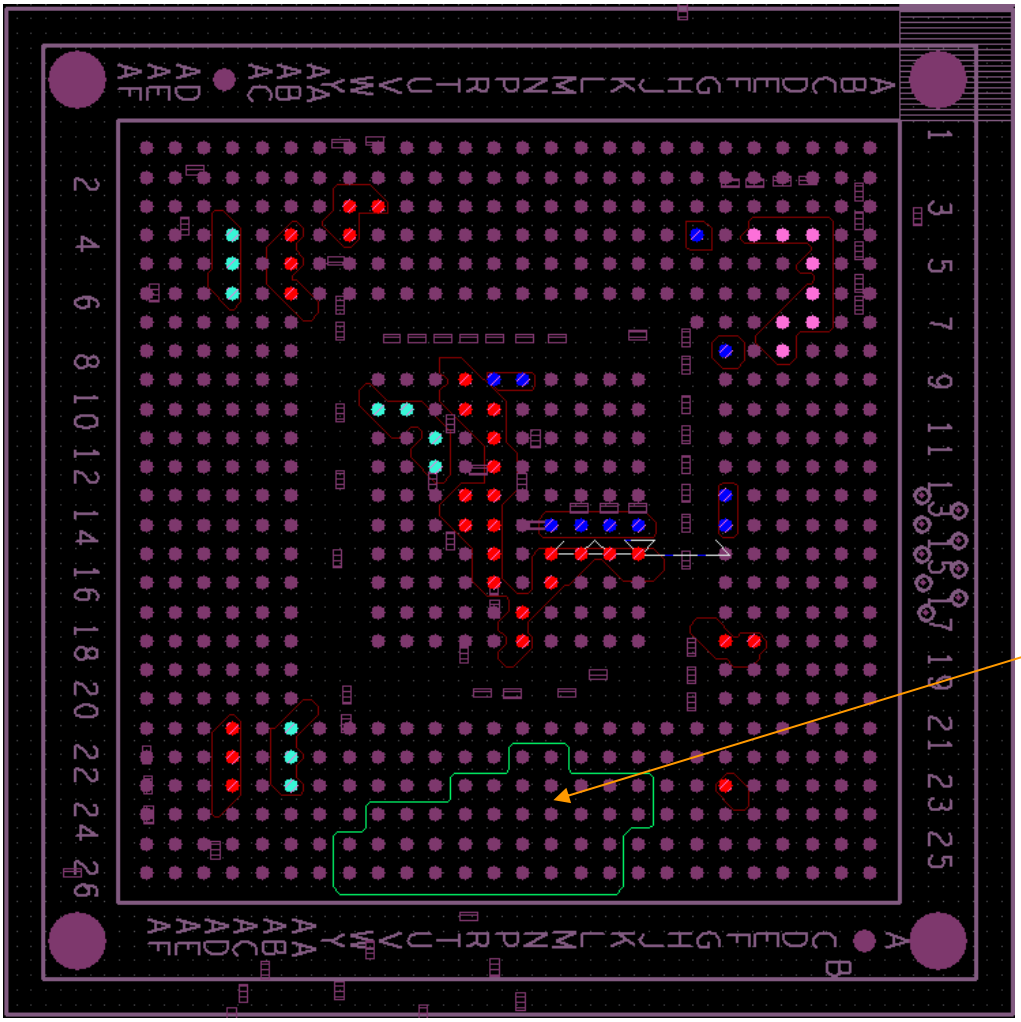
11. Routing Vref as short as possible with minimum 30mil trace width and at least 30mil space from other

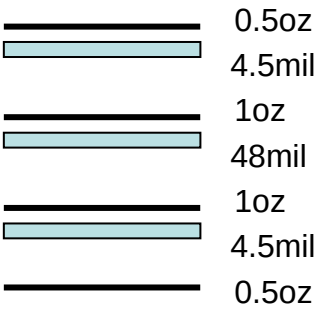
signal.

12. Matching trace length by serpentine line of “S” pattern. Make sure at least 5 times the distance of the trace to reference layer between line segments of “S” that parallel for the application of microstrip.
13. We are suggestion placed a bypass capacitor near vias every 4 to 5 vias for signal transition reference plane.
14. Place the resistors of bootstrap within 0.2 inch the pins of DDR2 device.

1.9. Analog area Layout Guideline

Avoid any digital signal cross the area. The constraint area is as follows:



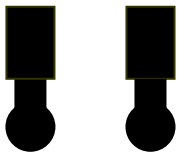


1.11. Bypass Capacitor scheme

The role of the bypass capacitor is to decouple the power supply bus from the IC. Bypass a power bus from the device internal noise requires capacitor with very low inductance. The objective is to eliminate the effects of the power bus inductance and resistance so that transient currents flowing across the power bus do not cause excessive noise at the power and ground pins of the IC. Therefore, the bypass capacitor should have low effective series resistance (ESR) and series inductance (ESL) while having a large enough capacitance value to supply current to the IC during switching.

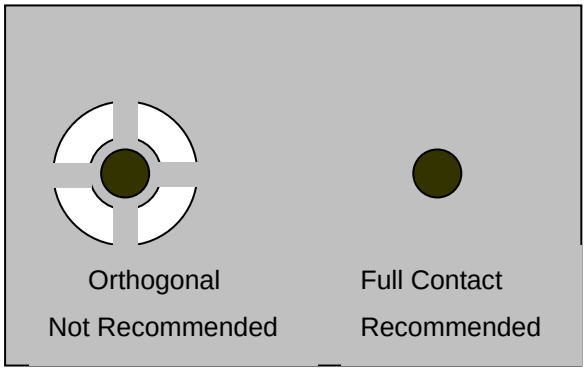
The recommendation bypass scheme for the large surge current is as follows:

1. It is recommended that at least a via per power pin for SDRAM memory interface
2. It is recommended that multiple vias per bypass capacitor to further reduce inductance.
3. It is recommended that X7R or X5R type and 0402 size capacitor for bypass.
4. It is recommended that a 47uf bulk capacitor for the interface.
5. The land pattern of bypass capacitor is as follows:



1.12. Suggestion Via for DDR2 and VORTEX86DX

In order to minimum the series inductance, ground and power via should be maximized as possible. We are suggestion on DDR deice that via is at least 18 mil pads with 10mil drilling size. We are also suggestion on A9120/A9121 device that via is at least 20 mil pads with 10mil drilling size. The via pattern connect to power/ground plan is as follows:



1.13. The DDR2 Vendors of pass DMP test program

Vendor	Density	Part number
NANYA	1G bits	NT5TU128M8DE-AC
Elpida	1G bits	E1108ACBG-8E-E
ASint	1G bits	DDRII1208-GE-I
Elpida	2G bits	E2108ABSE-6E-E